Image Operations

Negative ($s = L-1-r$): Inverts all intensities. Dark becomes bright, bright becomes dark.
Log Transform ($s = c\log(1+r)$): Compresses high intensities, expands low ones. Reveals detail in dark regions.
Power-Law / Gamma ($s = cr^{\gamma}$): $\gamma < 1$: brightens dark regions. $\gamma > 1$: darkens image, suppresses low intensities.
Contrast Stretching: Maps a narrow intensity range to the full $[0, L-1]$ range. Increases overall contrast.
Intensity-Level Slicing: Highlights a specific intensity band while suppressing or preserving all others.
Bit-Plane Slicing: Decomposes into 8 binary planes. Higher planes carry structure; lower planes carry fine detail/noise.
Thresholding ($s = 0$ or $L-1$): Produces binary image. Separates foreground from background.

Histogram Equalization: Flattens histogram to use full intensity range. Increases global contrast.
Histogram Matching: Reshapes histogram to match a specified target distribution.
Local Histogram Processing: Applies equalization in sliding neighborhoods. Reveals local detail that global methods miss.

Addition / Averaging: $\bar{g}(x,y) = \frac{1}{K}\sum g_i(x,y)$. Reduces noise variance by $1/K$. Images must be registered.
Subtraction: Highlights differences between two images. Removes static background.
Multiplication / Division: Mask with ROI to isolate regions. Division corrects uneven illumination/shading.

Smoothing (Box / Gaussian): Averages neighborhood pixels. Reduces noise but blurs edges.
Median Filter (nonlinear): Replaces pixel with neighborhood median. Removes salt-and-pepper noise while preserving edges.
Sharpening (Laplacian, Unsharp Mask): Enhances edges and fine detail. Makes transitions crisper.

NOT: swaps foreground/background. **AND**: intersection of two masks. **OR**: union of two masks. **XOR**: pixels in one mask but not both.

Image Overlap

$\cup$ = OR $\cap$ = AND A^c = NOT

Filter Kernels

Smoothing (Low-Pass)

Box: $\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\1 & 1 & 1\\1 & 1 & 1\end{bmatrix}$

Weighted Avg: $\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\2 & 4 & 2\\1 & 2 & 1\end{bmatrix}$

Gaussian: $h(x,y) = \frac{1}{2\pi\sigma^2}e^{-\frac{x^2+y^2}{2\sigma^2}}$

Sharpening (High-Pass)

Laplacian: $\begin{bmatrix}0 & 1 & 0\\1 & -4 & 1\\0 & 1 & 0\end{bmatrix}$

Laplacian (diag): $\begin{bmatrix}1 & 1 & 1\\1 & -8 & 1\\1 & 1 & 1\end{bmatrix}$

Unsharp (k=1): $\frac{1}{9}\begin{bmatrix}-1 & -1 & -1\\-1 & 17 & -1\\-1 & -1 & -1\end{bmatrix}$

Edge Detection (Gradient)

Sobel G_x: $\begin{bmatrix}-1 & 0 & 1\\-2 & 0 & 2\\-1 & 0 & 1\end{bmatrix}$

Sobel G_y: $\begin{bmatrix}-1 & -2 & -1\\0 & 0 & 0\\1 & 2 & 1\end{bmatrix}$

Prewitt G_x: $\begin{bmatrix}-1 & 0 & 1\\-1 & 0 & 1\\-1 & 0 & 1\end{bmatrix}$

Fourier Transform

Forward (spatial $\to$ frequency):

$$F(\mu) = \int_{-\infty}^{\infty} f(t)\, e^{-j2\pi\mu t}\, dt$$

Inverse (frequency $\to$ spatial):

$$f(t) = \int_{-\infty}^{\infty} F(\mu)\, e^{j2\pi\mu t}\, d\mu$$

Constant h over $[a,b]$:

$$\int_a^b h\, e^{-j2\pi ut}\, dt = \frac{h}{j2\pi u}\left(e^{-j2\pi ua} - e^{-j2\pi ub}\right)$$

Linear vs Nonlinear

Linear: output is weighted sum of inputs. Obeys superposition: $T[af_1 + bf_2] = aT[f_1] + bT[f_2]$. Examples: box filter, Gaussian, Laplacian.
Nonlinear: does not obey superposition. Examples: median filter, min/max filter.